ROS2 Robot Arm– Sorting Colored Blocks

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CSci 4551 • Introduction to Robotics

What's the problem? Why is it relevant to the course?

Sense

An overhead Depth camera captures color and depth frames of the workspace, identifying each object and estimating its 3D position and orientation on the table

Think

The system classifies each object by color, selects the matching bin, computes a grasp pose from the depth data, and plans a collision-free trajectory through pre-grasp, grasp, lift, transport, and drop

Act

A 6-DOF arm in Gazebo executes the planned trajectory, closes the gripper on the object, moves it to the correct bin, and releases

Project Outline

Camera / Perception	Cube + Basket Poses	Motion Planning & IK	Joint Trajectory + Gripper Plan	Arm Actuation

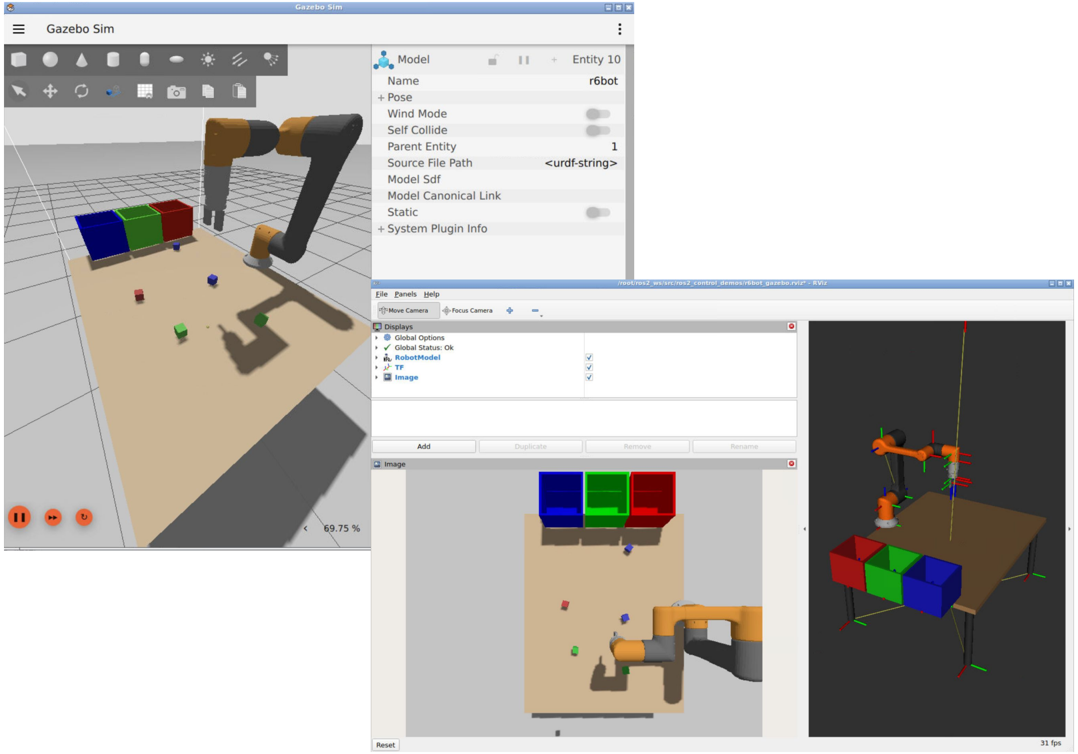
1. The perception node identifies cube positions and orientations via color segmentation and contour filtering in OpenCV, then converts pixel coordinates to the world frame.
2. A mission controller subscribes to detections, determines the pick-and-place order (matching each cube's color to the correct bin), and sends goal poses to MoveIt for each pick and place action.
3. MoveIt handles motion planning and execution — computing collision-free trajectories to the grasp pose, executing the pick with a fixed top-down approach, then planning and executing the trajectory to the target bin for release.

Progress so far, how is the project divided

- 6DOF robotic arm in gazebo
- Basic Ros system structure implemented
- Working perception node that can classify cube position and orientation
- Camara in Gazeabo that can see the table surface

Shawn: ROS2 system, getting environment working in gazebo, perception of cub position and orientation

Stanley: arm movement, cube grasping, and dropping into the matching bin



My Current Work: Arm Movement, Grasp, and Drop

This is the main manipulation pipeline I am implementing now

1. Receive target pose

Read cube position and matching basket position from the perception output

2. Move to pre-grasp

Plan a safe approach pose above the cube

3. Grasp the cube

Lower the arm, close the gripper, and confirm the object is held

4. Lift and transport

Raise the cube and move toward the correct basket

5. Release in bin

Open the gripper above the basket and reset for the next cube

Key robotics ideas used here

Forward kinematics to reason about end-effector pose

Inverse kinematics / waypoint generation to reach the cube and basket

Trajectory execution through ROS2 controllers

Frame consistency between camera coordinates and robot coordinates

Add arm trajectory / grasp screenshot

What We Will Do Next

Immediate next steps

- Implement mission controller to sequence pick-and-place
- Build arm controller with basic grasp planning
- Handle arm occlusion in perception
- Switch perception camara to world transform to TF2

After that

- Improve grasp reliability and success rate
- Tune pre-grasp approach and drop-off poses
- Extend to all cube colors
- Add obstacle avoidance
- Add retry logic for missed or failed grasps

Further

- Considering Blocks Overlap
- Sorting other shapes of objects
- Solving more complex tasks, like Tower of Hanoi
- Picking up cubes that are moving on a conveyor belt